

Classification of materials and their characteristics

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Solid materials have been conveniently grouped into three basic categories which are metal, polymer and ceramics, a science based primarily on chemical makeup and atomic structure.

metal - Normally metallic materials are combination of metallic elements. Metallic materials have large number of non-localized electrons, i.e., electrons are bound to particular atoms. Many properties of metals are directly attributable to these electrons.

* All metals are characterized by metallic properties e.g., luster, opacity, malleability, ductility and electrical conductivity.

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* Typical examples of metallic materials are iron, aluminium, copper, zinc etc and their alloys. They can be either bulk or powder form.

* Metals conduct electricity and heat, but don't transfer visible light. Moreover, metals are quite strong yet deformable.

Characteristics of metals

- luster
- hardness
- low specific heat
- good thermal conductivity
- good electrical conductivity

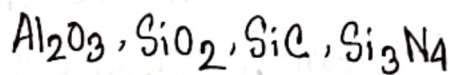
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- plastic deformability
- relatively high melting point
- strength
- ductility
- stiffness
- rigidity
- machinability
- weld ability
- cast ability.

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ceramics - ceramics are compound between metallic and non metallic elements. They are most frequently oxides, nitrides and carbides.



Traditional ceramics - clay mineral

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examples - porcelain, cements, glass

* ceramics are relatively stiff and strong in compare to any metal. Metals are ductile and ceramics are brittle.

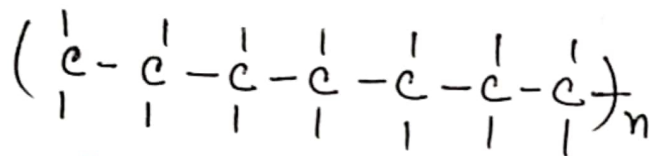
characteristics of ceramics

- brittleness
- rock-like stiffness
- resistance to high temperature

- hardness
- abrasiveness
- insulation (to follow of electric current)
- corrosion resistance
- opaque to light
- high temperature strength

Polymers: Polymer included the familiar plastic and rubber materials. Many of them are organic compound that are chemically based on carbon, hydrogen and other non-metallic elements. (O, N and Si)

• Furthermore, they have very large molecular structure, often chainlike structure/nature and backbone made of carbon atoms.



characteristics of polymer

- low weight
- combustible
- soft
- ductile
- not dimensionally stable
- poor conductivity
- poor resistance to temperature

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Questions

* How do you differ materials science from materials engineering?

Ans:

Materials science is the discipline of science which is primarily concerned with the search for basic knowledge about the material structures, properties and processing of materials.

Materials engineering is mainly concerned with the use of fundamental and applied knowledge of materials so that the materials can be converted into products necessary or desired by the society.

* Describe the role of materials in our daily life.

Ans.

1. Transportation
2. Housing
3. Clothing
4. Communication
5. Reaction
6. Food production
- 7.

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* Explain the selection criteria of a material for engineering applications.

Ans:

The choice of materials involves consideration of a number of factors which can be grouped under the following headings.

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1. Service requirement:

Strength: In most applications, some consideration must be given to the strength of the materials. For example, for gas transformation, the cylinder material should be strong enough to protect burst on due to high internal pressure.

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Corrosion Resistance: Many service environment are hostile to materials and chemically attack the surface. This attack takes a variety of forms which are grouped under the general little corrosion.

Electrical properties: Sometimes a major consideration in the selection of a material is the way it behaves in an electric circuit. It is necessary to know if it is a conductor or an insulator.

(ii) Manufacturing requirements: At the design stage, it is all easy to concentrate selecting the best combination of properties and ignore the fact that material must be shaped formed or joined. The most important are:

1. The material easy to machine.
2. Easy casting or molding
3. It can be welded or brazed successfully.

(iii) Cost of raw or bulk materials: Clearly the cost of raw material for the component of fabrication is of great importance. Depending on the type of product, the cost of materials may be as much as 40% of the total manufacturing cost.

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(iv) Customer preference: Finally the engineers must not forget that the customer may impose specific conditions which have a major influence on the choice of the most suitable material. For example,

* What are the scopes of materials science?

Scope of materials science

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- (i) Production of materials use in USA automobiles in the year 1992-2000
- (ii) Breakdown of weight percentages of major materials in the average 1985 automobiles. Some prediction have made in the past were of because the forestry to account for downsizing aerodynamics of use computer control of engine functions.
- (iii) Materials and materials process over the past year have been associated with increase in the performance of gas turbine engine
- (iv) The aircraft turbine engine is made principally of metal alloys. The latest high temperature-resistant, high strength nickel base alloys are use in this engine.
- (v) The success of the space plane project with depend on the development of new materials that can withstand externally high temperature.
- (vi) Potential ceramic components applications in a turbo charged diesel engine.

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(vii) Materials science based on the physics and chemistry of the internal structures of materials.

(viii) Investigates relations existing between the structure of materials and their properties.

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